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Empirical Research

Changes in Patient Care Experiences and the Nurse Work Environment: A Longitudinal Study of U.S. Hospitals

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Abstract

Addressing patient experience is a priority in the health care system. Hospital Consumer Assessment of Providers and Systems (HCAHPS) survey results incentivize hospitals to elevate patient experience, a factor in patient-centered care. Although hospital nursing resources have been positively associated with better HCAHPS ratings, it is unknown how changes in nursing resources are associated with changes in HCAHPS ratings over time. This two-period longitudinal study ranked the associations between changes in nurse staffing, skill mix, nurse education, and work environment on HCAHPS ratings and found that changes in the work environment had the strongest associations ($\beta = 2.29$; $p < .001$) with improved HCAHPS ratings. Our findings provide hospital administrators with empirical evidence that may help make informed decisions on how to best invest limited resources to improve HCAHPS ratings, including the potential utility of improving the work environment through enhancing Nursing Quality of Care and Nurse Participation in Hospital Affairs.

Keywords

HCAHPS, work environment, nursing, patient experience, patient satisfaction, staffing

Introduction

Patient-centered care, and by extension, patient experience, is at the forefront of quality health care. To improve these two aspects of health care, the Centers for Medicare and Medicaid Services (CMS) utilizes Hospital Consumer Assessment of Healthcare Providers and Systems (HCAHPS) surveying to measure and track hospitals' performances on patient experience. The program has financial incentives for hospitals to achieve "top-box" (i.e., a rating of 9 or 10 out of 10, most desirable) ratings. Therefore, hospital executives should be cognizant of modifiable organizational factors associated with "top-box" patient experience ratings to make evidence-based decisions on how to best utilize scarce resources to improve these ratings and maximize the potential for financial gain.

As the predominant health care provider at the bedside providing 24/7 care, nurses are vital to a patient's care experience. Nurse staffing levels and work environments are under hospital executives' purviews and are modifiable. Therefore, we propose that nursing resources such as whether there are enough registered nurses (RNs) at the bedside (i.e., staffing), whether the mix of nursing staff has a sufficient fraction of RNs (i.e., skill mix), what the educational composition of the nurses is, particularly nurses with a bachelor's degree or higher (i.e., nurse education; BSN), and whether the environment in which nurses work encourages autonomy and fosters positive nurse–physician relationships (i.e., work environment) are modifiable organizational factors that hospital administrators can influence. A growing body of research has explored whether modifiable nursing factors relate to patients' care experiences and has found statistically significant linkages between nursing resources and patient experience ratings in cross-sectional (Aiken et al., 2017; Boonpracom et al., 2018; Brooks Carthon et al., 2020; Chen et al., 2014; Kutney-Lee et al., 2009; Martsolf et al., 2016; Oppel & Young, 2018; Rosenbaum et al., 2024; Shindul-Rothschild et al., 2017; You et al., 2013) and longitudinal

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data (Aiken et al., 2018; Al-Amin et al., 2018; Delhy et al., 2021; Martsolf et al., 2016; Oppel & Young, 2018; Zhu et al., 2018). Evidence shows there is a knowledge gap, however, regarding how changes in multiple nursing resources relate to changes in patient experiences over time. That is, there are no longitudinal studies that examine all four modifiable nursing resources simultaneously and utilize standardized measurements to enable the comparison of effect sizes across nursing resources.

The purposes of this study were to: (a) identify whether and to what degree these nursing resources and patient experience ratings changed over time between 2006 and 2016; (b) determine the association between changes in nursing resources and changes in patient experience ratings; (c) identify which changes in nursing resources, comparatively, have the greatest associations with changes in patient experience ratings; and (d) determine whether changes in nursing resources over time have a greater association with patient experience ratings for hospitals with the lowest baseline patient experience ratings as compared with hospitals having the highest ratings in 2006. Our study's findings inform hospital executives regarding the degree to which changes in nursing resources may improve patient experience ratings.

New Contribution

To our knowledge, this is the first study statistically evaluating changes in multiple nursing resources simultaneously to determine which nursing resources have the greatest association with HCAHPS and, therefore, the potential to improve HCAHPS ratings. Determining that changes in nursing resources are associated with improved HCAHPS ratings and identifying which resources have the greatest association with changes in HCAHPS ratings provides hospital executives with the needed information to make evidence-based decisions when deciding how to spend limited resources.

Conceptual Framework

We conceptualize nursing resources as modifiable hospital characteristics (system attributes) that influence patient experience. The Quality Health Outcomes Model theorizes that the role of interventions, such as patient-centered nursing care or patient-centric nursing care activities and processes, are mediated via client (patient) and system attributes (Mitchell et al., 1998). This suggests there are no direct associations between patient-centered nursing care (interventions) and patient experience ratings (outcomes) rather that nursing resources (system attributes) are the link between interventions and outcomes. Although we do not directly measure patient-centered nursing care in this study, and because nursing resources enable nurses to provide quality care by giving them necessary resources to support the provision of patient-centered care, we hypothesize that nursing resources mediate the association between patient-centered nursing care and patient experience. For example, when nurses take care of fewer patients at a time, they have more time to spend with each patient. With sufficient staffing, nurses can provide critical surveillance that allows them to be responsive to patient's needs and not miss important nursing care tasks. Research shows that when nurses work in more supportive work environments (i.e., better work environments) they less often miss important care tasks like oral hygiene, skincare, treatments and procedures, and medication administration (Lake, Riman, & Sloane, 2020; Lake, Staiger, et al., 2020), and patients give higher patient experience ratings (Riman et al., 2022). Research supports that when hospitals employ a larger percentage of BSN-prepared nurses, patients experience better outcomes: lower odds of mortality, 7-day readmission, 30-day readmission, and shorter lengths of stay (Lasater et al., 2021; Porat-Dahlerbruch et al., 2022). We believe this is related to BSN-educated nurses having additional knowledge and skills related to better team coordination, critical decision-making, and patient advocacy. A skill mix that is richer in RNs (i.e., has a higher proportion of RNs to all nursing staff) is expected to expedite patient care because of their scope of decision-making and training. RNs have greater autonomy and authority to address patient concerns and/or needs and to do so on time when they have access to adequate nursing resources. As our focus was to examine the associations among nursing resources and patient experience and those associations over time, our research query is well suited for the theoretical components of the Quality Health Outcomes Model.

Method

Study Design

This longitudinal two-time period panel study employed three data sources, each at two time periods, those that most closely aligned with the nurse survey data which was collected in 2006 and 2016: the Center for Health Outcomes and Policy Research's RN4CAST data on hospital nursing resources (2006/07, 2016); CMS's HCAHPS survey data of hospital-level patient experience ratings (2007/08, 2016); and the American Hospital Association's (AHA) Annual Survey of Hospitals data on hospital structural characteristics (2006, 2016) to examine the associations between nursing resources and patient experience measured by HCAHPS ratings over time. Datasets were linked using a common hospital ID.

Data Sources

The RN4CAST is a dataset of survey responses from tens of thousands of licensed registered nurses in four states: California, Florida, New Jersey, and Pennsylvania. Nurses were asked to participate by mail using their home addresses obtained from state licensure lists. In 2016 nurses could complete the survey via mail or electronically. A 40% and

30% random sample of nurses in 2006 and 2016, respectively, were surveyed.

The four states from which nurses were surveyed represent a broad geographic area including more than 20% of annual hospitalizations (Aiken et al., 2011), and the represented hospitals account for more than 20% of all U.S. acute-care hospitals (Aiken et al., 2018). In addition, these four states are relatively large states and representative of rural and metropolitan areas, teaching and non-teaching hospitals, and hospitals with low and high technology capabilities. The nurse response rate was 39% in 2006 (Kutney-Lee et al., 2013) and 26% in 2016 (Lasater et al., 2019). To evaluate potential non-response bias, a non-respondent survey was conducted during data collection, revealing no non-response bias in staffing or the nurse work environment. Furthermore, although response rates were moderately low at the nurse level, hospitals were the unit of analysis and we have data from 98% of hospitals in the sampling frame that have more than 250 beds (Lasater et al., 2019). No other datasets have data on the nursing variables of interest across as many hospitals as these data represent. Registered nurses reported on their unit's staffing, skill mix, work environment, and their own level of education. They also provided the name of the hospital where they were employed, allowing for the linking of nursing resource data to HCAHPS and AHA data.

HCAHPS survey data are provided by CMS. These data are publicly available on the Care Compare website—formerly Hospital Compare. Data are reported every quarter. Included hospitals are listed by name and Provider Number allowing linkage to other data. This downloaded dataset provides hospital-level data on each hospital reporting HCAHPS ratings in 2007/2008 and 2016. Mandated reporting began in 2008, that is, October 2007. Note, data collected between October 2007-September 2008 are referred to herein as "HCAHPS 2006 data" to align with the proximate RN4CAST survey data. Prior to public reporting HCAHPS ratings, the CMS makes patient-mix adjustments at the hospital level, to improve comparability across hospitals. They include survey format, self-reported health status, education, service line, age, emergency room (ER) admission source, and if the spoken primary language is not English.

The American Hospital Association (AHA) Annual Survey provided data on hospital structural characteristics such as bed size, teaching status, technology capabilities, and geographic location.

Sample

All non-federal acute care hospitals within the four states (CA, FL, NJ, and PA) which had publicly reported HCAHPS ratings, RN4CAST data, and AHA data at both time points were included. Psychiatric, rehabilitation, free-standing pediatric hospitals, and free-standing ERs were excluded as HCAHPS surveys are not conducted in these settings. All included hospitals had at least five staff nurse respondents as the aggregated response from this minimum per hospital provides a reliable hospital-level value (Lake et al., 2021; Lake, Staiger, et al., 2020; Marsden et al., 2006). As HCAHPS ratings represent an overall hospital rating, nurses in all units for which HCAHPS data were collected were included (i.e., adult medical/surgical, adult intensive care unit [ICU], adult oncology, maternity [labor and delivery and mother baby], operating room, post-anesthesia care unit, intermediate care, telemetry, transitional care, and long-term care).

Variables

Predictors. Nursing resource measures were aggregated to the hospital-level from nurse-level responses (Lasater et al., 2019) with interclass correlation coefficients (ICCs) ranging between 0.75 and 0.78. Variables were measured the same in both years. Each respondent was asked how many patients, RNs, licensed practical nurses (LPNs)/licensed vocational nurses (LVNs), and unlicensed care personnel (nursing technicians or aides) were on their unit during their last shift. The measure of staffing was derived by nurses reporting the number of patients and the number of RNs on their units on their last shift and then aggregated to create a simple average across nurses. Skill mix was calculated as the proportion of staff RNs to all nursing staff which includes RNs, LPNs/LVNs, and unlicensed care personnel. Nurse education was defined as the percentage of RNs who held a bachelor's degree or higher. The Practice Environment Scale of the Nursing Work Index (PES-NWI) was used to measure the nurse work environment. The instrument evaluates the presence of five qualities within the nurse's workplace: (a) "Nurse Participation in Hospital Affairs" (e.g., nurses are engaged in hospital affairs such as creating policies)," (b) "Nursing Foundations for Quality of Care" (e.g., dedication to high quality nursing practice), (c) "Nurse Manager Ability, Leadership, and Support of Nurses" (e.g., nurses experience accessible and supportive nurse managers), (d) "Staffing and Resource Adequacy" (e.g., nurses have the time and resources to provide quality nursing care, i.e., sufficient nurse staffing), and (e) "Collegial Nurse-Physician Relations" (e.g., positive collaboration between nurses and physicians; Lake, 2002). This tool is composed of 31 Likert-type scale questions asking nurses to report the degree to which the nurse feels each characteristic is present in their hospital with a possible range from 1 to 4. Available answers include: 4 = "strongly agree," 3 = "agree," 2 = "disagree," or 1 = "strongly disagree." Individual subscales were calculated by averaging nurse responses for each item comprising the subscale among nurses working in the same hospital. A hospital-level composite measure was calculated as the average of all subscales. Higher scores indicate better work environments.

For each nursing resource- staffing, skill mix, nurse education, nurse work environment- a change variable was calculated by subtracting each hospital's 2006 nursing resource value from their 2016 resource value. Therefore, the nursing

resource change variable was the numeric value of change in a nursing resource between 2006 and 2016. For interaction models, hospitals were classified into terciles based on the percentage of patients that gave a "top-box" HCAHPS rating in 2006. Three HCAHPS baseline (2006) performance categories were created: tercile 1 "highest," tercile 2 "middling," tercile 3 "lowest." Note that we tested quartiles and quintiles which yielded similar results. We decided to report terciles as highest, middling, and lowest as those classifications have an intuitive interpretation and an equal number of hospitals in each category.

Outcome. As Overall Hospital Rating reflects a hospital's overall performance and is one of the few HCAHPS measures included in calculating VBP scores it was selected as the outcome variable of interest. We used CMS's rating system which reports the percentage of patients who answered a 9 or 10, known as a "top-box" rating, to the HCAHPS question, "Using any number from 0 to 10, where 0 is the worst hospital possible and 10 is the best hospital possible, what number would you use to rate this hospital during your stay?." Thus, HCAHPS ratings range from 0% to 100%. If a hospital had an HCAHPS rating of 74% and experienced an increase of 2 percentage points they would now have 76% of patients surveyed rating their hospital as a 9 or 10 (i.e., giving a "top-box" rating). We refer to this variable as the HCAHPS rating. To create a change in the HCAHPS rating variable we subtracted a hospital's 2006 HCAHPS rating from their 2016 rating, thus producing a numeric value of change.

Covariates. Bed size was categorized as small (<100 beds), medium (101-250 beds), or large (>250 beds). Hospital teaching status was measured as the ratio of physician residents and fellows to beds: nonteaching (no residents or fellows), minor (≤1:4 residents/fellows per bed), or major (>1:4 residents/fellows per bed). Hospitals with high-technology capabilities were hospitals that could perform open heart surgery and/or major organ transplants. Geographic location (i.e., rural, urban), was determined by the Metropolitan Statistical Area. HCAHPS survey response rates—the CMS-reported proportion of returned surveys—have been found to be positively associated with higher HCAHPS ratings (Godden et al., 2019). Thus, the CMS-reported survey response rate was used as a covariate in regression modeling to control for potential non-response bias (Godden et al., 2019; Tyser et al., 2016). Controlling for the proportion of ICU RNs addressed the inherent differences in staffing ratios between ICUs and the other units (e.g., staffing nurse-to-patient ratios are usually one nurse to one or two patients in the ICU and frequently one nurse to four or more patients in other units) and was calculated as the number of ICU staff RNs divided by the total number of staff nurses. The 2006 nursing resource values and HCAHPS ratings were used as controls in adjusted models to account for where hospitals began prior to any change (i.e., hospitals with lower baseline values have more room to change compared to hospitals with higher baseline values).

Analysis. If nurses did not completely answer the question their response was excluded from analysis. The percentage of hospitals with missing characteristics data was 2.5%. Missing data were filled in by evaluating hospitals' websites or imputing the data for the characteristic of the year that had data. For example, as hospital characteristics tend to not change over time, if a hospital had data on geographic location in 2006 but not 2016, we used the value for 2006 as the value for 2016. We manually searched to identify missing hospital characteristics and imputed those we could not identify. Statistical analyses were performed using STATA 17.0 statistical software. To determine the variation across samples and over time regarding key variables, descriptive statistics such as counts, percentages, frequencies, standard deviations (*SD*), means, and ranges were calculated. To accurately compare associations across nursing resources, each nursing resource variable was standardized using t-score scaling- which was essential as each nursing resource had its own metric of measurement. The STATA command *std(var)* was used. To examine multicollinearity both Spearman correlation and variance inflation factor (VIF) tests were conducted. $p < .05$ was considered statistically significant.

To determine whether changes in nursing resources are associated with changes in patient experience ratings over time, between 2006 and 2016, in a forward-stepwise fashion, bivariate and multivariate change in base ordinary least squares (OLS) linear regression models were estimated. We tested if changes in hospital characteristics had an impact on regression output and found negligible differences. So, to be consistent, the 2006 baseline hospital characteristics were used.

To determine whether changes in nursing resources over time, between 2006 and 2016, have a greater association with patient experience ratings for hospitals with the lowest ratings as compared with hospitals having the highest ratings in 2006, we repeated our OLS models by adding an interaction term between the change in a nursing resource over time and the hospital's 2006 baseline HCAHPS rating performance category with lowest as the reference category. The estimation of all four resources in one model including interaction terms made the model statistically unstable, therefore, the model was estimated four times, once for each nursing resource.

We carried out post hoc analyses of the work environment subscales using OLS modeling without interaction terms and only in independent (unadjusted and adjusted) models due to subscales demonstrating high multicollinearity. We used standardized variables and *t*-scores to compare effect size and statistical significance between subscales. This was done to provide further details on which subscales may be the most predictive of high HCAHPS rating performance.

Results

The sample included survey responses from 30,512 RNs in 489 hospitals. These hospitals tended to be large (49% in 2006 and 54% in 2016), had high technology capabilities (53% in 2006 and 56% in 2016), and were located in urban areas (92% in 2006 and 94% in 2016). Eighty-one (17%) hospitals had a change in bed size with all hospitals becoming larger. Thirty-two percent ($n=157$) of hospitals had a change in teaching status with the most change (10%) being in non-teaching hospitals in 2006, becoming either minor or major teaching hospitals in 2016. Only 6% of hospitals had a change in technology capabilities (Table 1).

The mean change in the HCAHPS Overall Rating was +8.17 ($SD=7.63\%$) percentage points with a range between −15% and 38%. The mean change in staffing was −0.63 ($SD=1.18$), which was a desirable change on average since a reduction in staffing indicates fewer patients per nurse. The mean change in skill mix had favorable improvement over time (+6%; $SD=8.54\%$), as indicated by a 6% increase in the proportion of registered nurses providing direct patient care relative to other nursing staff. The mean change in hospitals' proportion of BSN-educated nurses also had a favorable improvement of +14% ($SD=17.17\%$) more BSN-educated nurses. Finally, the mean score of hospitals' nurse work environment improved by +0.06 on a scale of 1 to 4 ($SD=0.29$; Table 1).

Table 2 displays the results from OLS regression modeling estimating the associations of changes in nursing resources and the changes in the percentage of patients that gave "top-box" ratings between 2006 and 2016. The variable change in the work environment, in all models, had the largest standardized beta coefficients (2.29; model 4) and the smallest p values ($p<.001$) compared with the other three nursing resources and was the only resource that was statistically significantly associated with changes in patient experience over time. That is, 1 *SD* higher work environment rating was associated with an additional 2.29 percentage points higher of patients giving "top-box" ratings. This is a clinically meaningful increase in HCAHPS ratings and akin to a hospital moving from performing in the 50th percentile to the 62nd percentile compared to peer hospitals in improving their HCAHPS ratings. Change in skill mix had small beta coefficients and was never significant. For both change in staffing and change in nurse education, statistical significance varied across models. Change in staffing was significant in independent models and change in nurse education was never significant in an adjusted model (see Appendix A for unadjusted models). In adjusted models (Models 1 and 2), change in nurse education switched the direction of association to become positive with much smaller beta coefficients and lost statistical significance. In addition, within these adjusted models, change in staffing achieved the second largest standardized beta coefficients −1.16 (Model 1) and −0.47 (Model 2), but statistical significance ($p=.001$) was only achieved in the independent adjusted model (model 1). We identified that a 0.29-point improvement in the work environment over time was associated with a 2.29 increase ($p<.001$; model 2) in the percentage points of patients giving "top-box" ratings.

The only significant interaction was between change in work environment and hospital 2006 baseline HCAHPS rating performance category, specifically for the lowest vs. middling performance categories ($p=.037$; Figure 1). The presence of a significant interaction implies that the association of the change in the work environment is not the same for hospitals at every baseline performance category. A one *SD* (0.29) change in the work environment produced a 1.45 percentage point change in the percentage of patients giving "top-box" ratings in the hospitals in the lowest performing category, but a (1.45 + 1.45=2.9) or roughly 3 percentage point change for hospitals in the middling performance category, and a (1.45 + 0.47=1.92) or roughly 2 percentage point change for hospitals in the highest performing category. In a margins plot (Figure 1) we see that the line representing the average group has a steeper slope indicating that the association of change on HCAHPS ratings— in association with a one *SD* change in the work environment— is greatest in the middling group when compared with the lowest and compared with the lowest versus the highest.

Among the subscales of the work environment, changes in Collegial Nurse-Physician Relations changed the most ($M=+0.12$; $SD=0.29$) between 2006 and 2016 followed by Nurse Manager Ability, Leadership, and Support of Nurses ($M=+0.10$; $SD=0.38$), while Nurse Resource Adequacy decreased ($M=-0.03$; $SD=0.38$; Table 1). Over time, changes in all subscales were significantly associated with changes in HCAHPS ratings. Changes in Nursing Foundations for Quality of Care (standardized beta=2.26; t-score=8.28) and changes in Nurse Participation in Hospital Affairs (standardized beta=2.18; t-score=8.24) were associated with the greatest amount of change in HCAHPS ratings in adjusted linear regression modeling (Figure 2).

Discussion

In a study examining changes in nursing resources and HCAHPS ratings over a decade, we found that all nursing resources changed over time with various magnitudes of change and that changes in these resources were associated with changes in HCAHPS ratings. We identified that change in the work environment had statistically significant associations with higher HCAHPS ratings within all models.

Two specific aspects of the work environment, change in Nursing Foundations for Quality of Care (beta coefficient 2.26) and change in Nurse Participation in Hospital Affairs (beta coefficient 2.18), were determined to have the most association with improving HCAHPS ratings. The other three domains had beta coefficients as low as 1.86 (change in Collegial Nurse-Physician Relationships) and as high as 2.16

Table 1. Changes Over Time in Key Variables Between 2006 and 2016.

	2006: *n* (%)	2016: *n* (%)	Amount of change
Variable	*n* = 489	*n* = 489	*n* (%)
Hospital characteristics			
Bed size			81 (17%)
Small	38 (8%)	30 (6%)	
Medium	212 (43%)	197 (40%)	
Large	239 (49%)	262 (54%)	
Teaching status			157 (32%)
None	243 (50%)	198 (40%)	
Minor	205 (42%)	243 (50%)	
Major	41 (8%)	48 (10%)	
Technology capabilities			31 (6%)
Low	228 (47%)	213 (44%)	
High	261 (53%)	276 (56%)	
Geographic location			11 (2%)
Urban	450 (92%)	461 (94%)	
Rural	39 (8%)	28 (6%)	
State			0 (0%)
California	191 (39%)	191 (39%)	
Florida	135 (28%)	135 (28%)	
New Jersey	57 (12%)	57 (12%)	
Pennsylvania	106 (22%)	106 (22%)	
Outcome			
Overall HCAHPS Rating			
M	59.78%	67.95%	8.17%
SD	8.86%	7.39%	7.63%
Range	30% to 91%	42% to 90%	−15% to 38%
Nursing resources			
Staffing			
M	4.69	4.05	−0.63
SD	1.08	0.98	1.18
Range	1.79 to 8.38	1.76 to 10.06	−5.71 to 3.58
Skill mix			
M	75.36%	80.86%	5.50%
SD	7.34%	6.81%	8.54%
Range	46.27% to 100%	59.66% to 100%	−22.12% to 44.73%
Education			
M	40.67%	54.71%	14.04%
SD	14.51%	17.82%	17.17%
Range	0% to 84.62%	0% to 100%	−67.78% to 66.67%
Work environment			
M	2.70	2.77	0.06
SD	0.23	0.29	0.29
Range	2.03 to 3.27	1.71 to 3.59	−0.86 to 0.07
PES-NWI subscales			
Nursing foundations for quality of care			
M	2.96	3.04	0.08
SD	0.23	0.28	0.28
Range	2.26 to 3.57	2.1 to 3.74	−0.78 to 1.03
Nurse participation in hospital affairs			
M	2.58	2.62	0.05
SD	0.30	0.38	0.37
Range	1.55 to 3.37	1.52 to 3.54	−1.01 to 1.20

(continued)

Table 1. (continued)

Variable	2006: *n* (%)	2016: *n* (%)	Amount of change
	n = 489	*n* = 489	*n* (%)
Nurse manager ability, leadership, & support of nurses			
M	2.61	2.71	0.10
SD	0.28	0.35	0.38
Range	1.7 to 3.63	1.39 to 3.88	−1.25 to 1.47
Staffing & resource adequacy			
M	2.49	2.46	−0.03
SD	0.31	0.37	0.38
Range	1.58 to 3.29	1.35 to 3.5	−1.16 to 1.22
Collegial nurse-physician relations			
M	2.88	3.00	0.12
SD	0.23	0.28	0.29
Range	2.20 to 3.47	1.47 to 3.76	−1.09 to 0.92

Note. SD = standard deviation.

(change in Nurse Manager Ability, Leadership, and Support of Nurses; see Figure 2). When examining the interactions between a hospital's baseline HCAHPS performance category, change in work environment had a higher association with HCAHPS ratings for hospitals in the middling baseline performance category as compared with those in the highest or lowest categories.

Improving patient experience is of value to hospitals striving to provide quality, patient-centered care. Enhancing HCAHPS ratings may allow hospitals to increase their patients' care experience ratings and maximize financial incentives from the CMS. Therefore, hospital executives are keen to determine evidence-based interventions to improve patient experience. Change in the work environment appears to be an identified lever to achieve the desired higher HCAHPS ratings.

We found that the change in staffing over time was significant, but only in models where only one nursing resource at a time was examined. This finding may be related to Aiken and colleague's (2011) finding that the benefits of good staffing are pronounced in hospitals with the best work environments and diminished in those with poorer work environments. This is to say, that investments in both staffing and the work environment in tandem may exert a synergistic effect on hospitals' abilities to obtain a higher proportion of "top-box" ratings.

Our work builds upon that of Aiken and colleagues (2018) who conducted the only prior study to examine changes in the work environment and patient experience ratings longitudinally, in a hospital panel equivalent to the current study. They linked a 10% or greater change in the work environment to changes in Overall Rating. The present study adds nuance about changes in the work environment and changes in HCAHPS ratings and shows that, of the four nursing resources, the work environment is the most predictive of changes in high patient experience ratings. Nuances unique to our study include: (a) evaluation of more granular change in the work environment (e.g., changes less than 10%); (b) comparison associations of changes in multiple nursing resources simultaneously; and (c) examination of interactions between changes in the work environment and hospital baseline performance category. The addition of our more specific findings may enable hospital executives to identify: (a) what their hospital's performance category is and (b) how much they could anticipate needing to improve their work environment to achieve their targeted HCAHPS ratings and performance category.

A novel contribution of our study was the determination of how a hospital's baseline performance relates to its potential improvement in patient experience ratings. We found that only the change in the work environment had a statistically significant interaction with a hospital's baseline performance category. Here, a change of one *SD* in the change in work environment yielded a greater change in HCAHPS ratings for those hospitals that were in the middling category as compared to those in the lowest category at baseline. The explanation for this finding is not clear if we suppose that hospitals that began with the fewest resources (i.e., those in the lowest category at baseline) would be most sensitive to work environment improvements. Perhaps hospitals in the middling category had a prerequisite work environment floor/foundation to which improvements could translate into better patient experiences, as compared with those in the lowest category, while simultaneously not being constrained by the ceiling effect of those in the highest category. We observed that hospitals in the middling category exhibited the largest work environment improvement which may reveal this greater improvement capacity among this hospital subgroup.

Our goal was to select the top two areas of the work environment that hospital administrators could invest in to gain greater returns on investments. First, we identified Nursing

Table 2. Effects of Changes in Nursing Resources on Changes in HCAHPS Rating.

	Model 1 (Joint unadjusted)		Model 2 (Joint adjusted)	
Variable	Beta coefficient [95% CI]	*p*-value	Beta coefficient [95% CI]	*p*-value
Change in staffing	−0.52 [−1.25, 0.21]	.165	−0.47 [−1.15, 0.20]	.171
Change in skill mix	−0.10 [−0.80, 0.61]	.790	0.16 [−0.49, 0.81]	.627
Change in education	−1.11 [−1.78, −0.45]	.001	0.10 [−0.46, 0.67]	.721
Change in work environment	1.33 [0.64, 2.02]	<.001	2.29 [1.75, 2.83]	<.001

Note. Standardized beta coefficients represent the expected change in HCAHPS ratings per an increase of one standard deviation (*SD*) increase each nursing resource. Nursing resource (*SD*) as follows: change in staffing (1.18; 1 additional patient per nurse), change in skill mix (8.54% increase in the proportion of RNs to total nursing staff), change in education (17.17% increase in the percentage of BSN prepared RNs), and change in work environment (0.29 increase on a 1-4 scale). Model 1: All four nursing resources are modeled jointly in an unadjusted model. Model 2: All four nursing resources are modeled jointly in an adjusted model. Models are adjusted for 2006 hospital characteristics: size, teaching status, technology capability, geographic location, state, HCAHPS survey response rate, and proportion of ICU RNs.

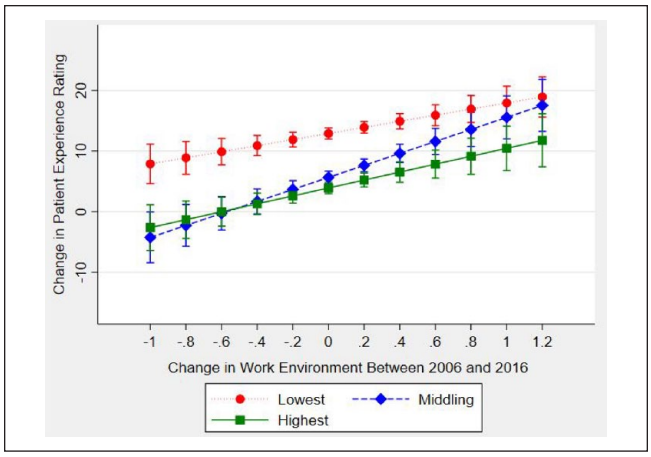

Figure 1. Margins Plot of Interaction Effects Between Changes in the Work Environment and 2006 Baseline Patient Experience Performance Category on the Changes in the Percentage of Patients That Gave "Top-Box" Overall HCAHPS Rating.
Note. The lowest is the reference category and vertical bars represent 95% confidence intervals. In an adjusted model: change in work environment main effect (standardized beta coefficient: 1.45; $p = .001$); main effect baseline middling performance category (beta coefficient: −7.44; $p \leq .001$); main effect baseline highest performance category (beta coefficient: −11.26; $p \leq .001$); interaction term between lowest and highest (standardized beta coefficient: 1.60; $p = .482$); and lowest and middling (standardized beta coefficient: 1.45; $p \leq .037$).

Foundations for Quality of Care which measures if nurses are provided with essential tools to provide quality nursing care. These tools include having preceptor programs for new graduates, high standards of what quality nursing care is, and different ways for nurses to continue learning (e.g., in unit education classes). The second area was Nurse Participation in Hospital Affairs which measures how engaged nurses are in hospital decision making. As the only health care providers at the bedside around the clock, nurses are a vital part of hospital affairs. Nurses should have the opportunity to advance their careers, have access to nursing administrators, and be able to participate in decision-making (e.g., policies and protocols). Our findings suggest that prioritizing improvement in these two areas of the work environment may have a more significant association with improved patient experience ratings. Notably, Lake (2002) asserted that these two aspects of the work environment reflect hospital-wide work environment domains, as compared with the other three subscales, which were considered to reflect nursing-unit-level domains. Our results may be interpreted as supporting that assertion. Hospital executives regularly surveying their own nurses regarding their current work environments and developing tailored interventions may see the most improvement in patient experience ratings.

We identified that the work environment was strongly associated with higher HCAHPS ratings, and that two aspects of the work environment yielded a greater association with patient experience ratings. We do not know, however, how to best make changes in the work environment to maximize patient experience ratings. Therefore, there is a need for future research on interventions that explicitly improve their work environment to equate to a 0.29-point change in the work environment. Research to make this determination would be beneficial to hospital executives striving to make evidence-based investment decisions. Although our work illuminates the potential that improvements in the work environment have for improving HCAHPS ratings, and therefore hospital financials, we did not explicitly examine those associations. Further research should be conducted to determine the associated financial benefits of improved HCAHPS ratings related to improved work environments.

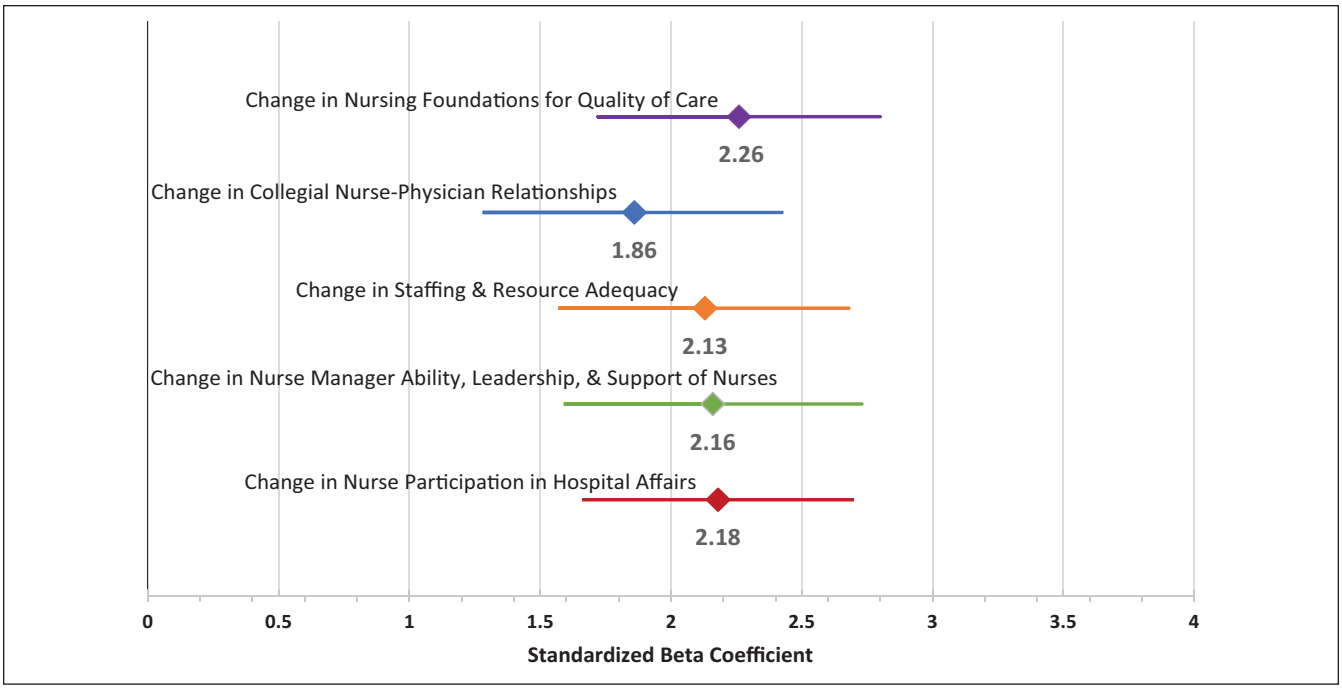

Figure 2. Comparison of Change in Work Environment Subscales Effect on Change in HCAHPS Rating.
Note. Standardized beta coefficients are from a standard linear regression model. The x-axis represents the standardized beta coefficients which are represented as diamonds on the graph. These beta coefficients represent the anticipated change in patient experience ratings for an increase of one *SD* in a given subscale. The colored horizontal lines represent the 95% confidence intervals. Subscale (*SD*; t-score) as follows: Change in Nursing Foundations for Quality of Care (0.28; 8.28), Change in Collegial Nurse-Physician Relationships (0.29; 6.34), Change in Staffing & Resource Adequacy (0.38; 7.55), Change in Nurse Manager Ability, Leadership, & Support of Nurses (0.38; 7.41), and Change in Nurse Participation in Hospital Affairs (0.37; 8.24). Models are adjusted for 2006 hospital characteristics: size, teaching status, technology capability, geographic location, state, HCAHPS survey response rate, and proportion of ICU RNs.

Although our data are no more recent than 2016, they are the most recent data in the panel of these four states. Our data are advantageous in providing insights over a decade to allow us to identify changes in HCAHPS ratings from the initiation of public reporting and the dates of our data are comparable to data used in current patient experience literature (Aiken et al., 2021; Brooks Carthon et al., 2020; Delhy et al., 2021; McCaughey et al., 2020; Riman et al., 2022; Rosenbaum et al., 2024). Our data are not nationally representative but do allow for some generalizability as they represent four large geographically diverse states in which 20% of all annual hospital discharges occur. Results, however, should be evaluated in the context of this study. HCAHPS data are only publicly available at the hospital level not at an individual level. This limits us in understanding differences among patients with varying sociodemographic backgrounds and clinical differences, which may also be differentially affected by nursing resources. Tercile cutoffs may be different within a different sample as such, replicability and generalizability may be affected. However, we believe that our large geographically diverse sample lessens this possibility. Based on two time points, and potential confounders that we were unable to measure in our data like incentives introduced with the Affordable Care Act or Medicaid expansion in some states, it is unclear how much of a straight-line association there is between these factors over time. There may be more complexity to the trends in HCAHPS performance than could be accounted for in the present study. We can, however, see that changes in nursing resources are concomitantly associated with HCAHPS ratings over time.

Conclusion

Improving patient experience is an important step toward achieving patient-centered care and enhancing VBP scores linked to hospital Medicare reimbursements. Data from 30,512 nurses in 489 hospitals substantiates that improving work environments and nurse staffing over time are organizational interventions that may result in more favorable patient experience ratings. Interventions targeting particular features of the work environment, specifically Nursing Foundations for Quality of Care (e.g., valuing and supporting quality nursing care) and Nurse Participation in Hospital Affairs (e.g., fostering nurses' participation in hospital decision-making), may present a mechanism for improving patient experience ratings.

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Supplemental Material

Supplemental material for this article is available online.

References

Aiken, L. H., Cimiotti, J. P., Sloane, D. M., Smith, H. L., Flynn, L., & Neff, D. F. (2011). Effects of nurse staffing and nurse education on patient deaths in hospitals with different nurse work environments. *Medical Care*, *49*(12), 1047–1053.

Aiken, L. H., Simonetti, M., Sloane, D. M., Cer'on, C., Soto, P., Bravo, D., Galiano, A., Behrman, J. R., Smith, H. L., McHugh, M. D., & Lake, E. T. (2021). Hospital nurse staffing and patient outcomes in Chile: A multilevel cross-sectional study. *The Lancet*, *9*, e1145–e1153. https://doi.org/10.1016/S2214-109X(21)00209-6

Aiken, L. H., Sloane, D., Griffiths, P., Rafferty, A. M., Bruyneel, L., McHugh, M., Maier, C. B., Moreno-Casbas, T., Ball, J. E., Ausserhofer, D., Sermeus, W., Heede, K. V. D., Lesaffre, E., Diya, L., Smith, H., Jones, S., Kinnunen, J., Ensio, A., Jylhä, V., . . .Van Achterberg, T. (2017). Nursing skill mix in European hospitals: Cross-sectional study of the association with mortality, patient ratings, and quality of care. *BMJ Quality and Safety*, *26*(7), 559–568. https://doi.org/10.1136/bmjqs-2016-005567

Aiken, L. H., Sloane, D. M., Barnes, H., Cimiotti, J. P., Jarrín, O. F., & McHugh, M. D. (2018). Nurses' and patients' appraisals show patient safety in hospitals remains a concern. *Health Affairs*, *37*(11), 1744–1751. https://doi.org/10.1377/hlthaff.2018.0711

Al-Amin, M., Schiaffino, M. K., Park, S., & Harman, J. (2018). Sustained hospital performance on hospital Consumer Assessment of Healthcare Providers and Systems survey measures: What are the determinants? *Journal of Healthcare Management*, *63*(1), 15–28. https://doi.org/10.1097/JHM-D-16-00006

Boonpracom, R., Kunaviktikul, W., Thungjaroenkul, P., & Wichaikhum, O. (2018). A causal model for the quality of nursing care in Thailand. *International Nursing Review*, *66*(1), 130–138. https://doi.org/10.1111/inr.12474

Brooks Carthon, J. M., Hatfield, L., Brom, H., Houton, M., Kelly-Hellyer, E., Schlak, A., & Aiken, L. H. (2020). System-level improvements in work environments lead to lower nurse burnout and higher patient satisfaction. *Journal of Nursing Care Quality*, *36*(1), 7–13. https://doi.org/10.1097/NCQ.0000000000000475

Chen, J., Koren Laine, M. E., Munroe, D. J., & Yao, P. (2014). Is the hospital's magnet status linked to HCAHPS scores? *Journal of Nursing Care Quality*, *29*(4), 327–335. https://doi.org/10.1097/NCQ.0000000000000062

Delhy, R., Dor, A., & Pittman, P. (2021). The impact of nursing staff on satisfaction scores for U.S. hospitals: A production function approach. *Medical Care Research and Review*, *78*(6), 672–683. https://doi.org/10.1177/1077558720950572

Godden, E., Paseka, A., Gnida, J., & Inguanzo, J. (2019). The impact of response rate on Hospital Consumer Assessment of Healthcare Providers and System (HCAHPS) dimension scores. *Patient Experience Journal*, *6*(1), 105–114. https://doi.org/10.35680/2372-0247.1357

Kutney-Lee, A., McHugh, M. D., Sloane, D. M., Jeannie, P., Flynn, L., Neff, D. F., & Aiken, L. H. (2009). Nursing: A key to patient satisfaction. *Health Affairs*, *28*(Suppl. 3), 669–677. https://doi.org/10.1377/hlthaff.28.4.w669

Kutney-Lee, A., Sloane, D. M., & Aiken, L. H. (2013). An increase in the number of nurses with baccalaureate degrees is linked to lower rates of postsurgery mortality. *Health Affairs*, *32*(3), 579–586.

Lake, E. T. (2002). Development of the practice environment scale of the nursing work index. *Research in Nursing and Health*, *25*(3), 176–188. https://doi.org/10.1002/nur.10032

Lake, E. T., French, R., Clark, R. R. S., O'Rourke, K., & Lorch, S. (2021). Newborns with neonatal abstinence syndrome are concentrated in poorer-quality hospitals. *Hospital Pediatrics*, *11*(4), 342–349. https://doi.org/10.1542/hpeds.2020-003145

Lake, E. T., Riman, K. A., & Sloane, D. M. (2020). Improved work environments and staffing lead to less missed nursing care: A panel study. *Journal of Nursing Management*, *28*(8), 2157–2165. https://doi.org/10.1111/jonm.12970

Lake, E. T., Staiger, D. O., Cramer, E., Hatfield, L. A., Smith, J. G., Kalisch, B. J., & Rogowski, J. A. (2020). Association of patient acuity and missed nursing care in U.S. Neonatal Intensive Care Units. *Medical Care Research and Review*, *77*(5), 451–460. https://doi.org/10.1177/1077558718806743

Lasater, K. B., Aiken, L. H., McHugh, M. D., Sloane, D. M., Smith, H. L., Policy, H. C., & Brunswick, N. (2019). A methodology for studying organizational performance: A multistate survey of front-line providers. *Medical Care*, *57*(9), 742–749. https://doi.org/10.1097/MLR.0000000000001167.A

Lasater, K. B., Sloane, D. M., McHugh, M. D., & Aiken, L. H. (2021). Changes in proportion of bachelor' s nurses associated with improvements in patient outcomes. *Research in Nursing & Health*, *44*(5), 797–795. https://doi.org/10.1002/nur.22163

Marsden, P. V., Landon, B. E., Wilson, I. B., McInnes, K., Hirschhorn, L. R., Ding, L., & Cleary, P. D. (2006). The reliability of survey assessments of characteristics of medical clinics. *Health Services Research*, *41*(1), 265–283. https://doi.org/10.1111/j.1475-6773.2005.00480.x

Martsolf, G. R., Gibson, T. B., Benevent, R., Jiang, H. J., Stocks, C., Ehrlich, E. D., Kandrack, R., & Auerbach, D. I. (2016). An examination of hospital nurse staffing and patient experience with care: Differences between cross-sectional and longitudinal estimates. *Health Services Research*, *51*(6), 2221–2241. https://doi.org/10.1111/1475-6773.12462

McCaughey, D., McGhan, G. E., Rathert, C., Williams, J. H., & Hearld, K. R. (2020). Magnetic work environments: Patient experience outcomes in Magnet versus non-Magnet hospitals. *Health Care Management Review*, *45*(1), 21–31. https://doi.org/10.1097/HMR.0000000000000198

Mitchell, P. H., Ferketich, S., & Jennings, B. M. (1998). Quality health outcomes model. *Image: The Journal of Nursing Scholarship*, *30*(1), 43–46. https://doi.org/10.1111/j.1547-5069.1998.tb01234.x

Oppel, E. M., & Young, G. J. (2018). Nurse staffing patterns and patient experience of care: An empirical analysis of U.S. hospitals. *Health Services Research*, *53*(3), 1799–1818. https://doi.org/10.1111/1475-6773.12756

Porat-Dahlerbruch, J., Aiken, L. H., Lasater, K. B., Sloane, D. M., & McHugh, M. D. (2022). Variations in nursing baccalaureate education and 30-day inpatient surgical mortality. *Nursing Outlook*, *70*(2), 300–308. https://doi.org/10.1016/j.outlook.2021.09.009

Riman, K. A., Harrison, J. M., Sloane, D. M., & McHugh, M. D. (2022). Work environment and operational failures associated with nurse outcomes, patient safety, and patient satisfaction. *Nursing Research*, *72*(1), 20–29. https://doi.org/10.1097/nnr.0000000000000626

Rosenbaum, K. E. F., Lasater, K. B., McHugh, M. D., & Lake, E. T. (2024). Hospital performance on hospital consumer assessment of healthcare providers and system ratings: Associations with nursing factors. *Medical Care*, *62*(5), 288. https://doi.org/10.1097/MLR.0000000000001966

Shindul-Rothschild, J., Flanagan, J., Stamp, K. D., & Read, C. Y. (2017). Beyond the pain scale: Provider communication and staffing predictive of patients' satisfaction with pain control. *Pain Management Nursing*, *18*(6), 401–409. https://doi.org/10.1016/j.pmn.2017.05.003

Tyser, A. R., Abtahi, A. M., McFadden, M., & Presson, A. P. (2016). Evidence of non-response bias in the Press-Ganey patient satisfaction survey. *BMC Health Services Research*, *16*(1), 350. https://doi.org/10.1186/s12913-016-1595-z

You, L., Aiken, L. H., Sloane, D. M., Liu, K., He, G., Hu, Y., Jiang, X., Li, X., Li, X., Liu, H., Shang, S., Kutney-Lee, A., & Sermeus, W. (2013). Hospital nursing, care quality, and patient satisfaction: Cross-sectional surveys of nurses and patients in hospitals in China and Europe. *International Journal of Nursing Studies*, *50*(2013), 154–161. https://doi.org/10.1016/j.ijnurstu.2012.05.003

Zhu, J., Dy, S. M., Wenzel, J., & Wu, A. W. (2018). Association of magnet status and nurse staffing with improvements in patient experience with hospital care, 2008-2015. *Medical Care*, *56*(2), 111–120. https://doi.org/10.1097/MLR.0000000000000854